



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2025-26
COURSE HANDOUT PART II

Date: 2.08.25

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No : CE F313
Course Title : Foundation Engineering
Instructor-in-Charge : Ravi Kant Mittal (ravimittal@pilani.bits-pilani.ac.in) Mb.9887692025
Instructor(s) : -
Tutorial Instructors : Ravi Kant Mittal, Rakshanda Showkat, Vijaykumar (RS), Harish Kuppili(RS)

1. Course Description: General requirement for satisfactory performance of shallow foundations, general, local and punching shear failures, bearing capacity, settlement, tilt and rotation of foundations, proportioning of Shallow Foundations, footings on layered soils and slopes, Deep foundations, capacity of single and group Piles, laterally loaded pile, Earth Pressure theories, Retaining structures, design and checks for stability, Stability of slopes, Introduction to Ground Improvement Techniques and geosynthetics with applications, Introduction to machine foundations for different type of machines, Introduction to geotechnical earthquake engineering and liquefaction of soils, computer applications in foundation design

2. Scope and Objective of the Course: The main goal of this course is to provide an in-depth understanding regarding different types of foundations. Analysis of foundations and retaining structures (Shallow foundations, pile foundations, machine foundations, retaining walls, slope stability etc.) considering all geotechnical aspects. Emphasis will be given on complete coverage of code of practices for various types of foundations and retaining structures. Geotechnical earthquake engineering, ground improvement techniques, reinforced earth walls and geosynthetics applications increased tremendously therefore given due consideration.

3. Course learning outcomes:

CLO1.Geotechnical design of Shallow and deep foundations

CLO2.Earth pressure computation, stability of retaining structure and slopes

CLO3.Introduction to emerging area of ground improvement, machine foundations and geotechnical earthquake engineering

4. Text Books and reference books: T1. Murthy, V. N. S. (2010) “Geotechnical Engineering: Principles and Practices of Soil Mechanics and Foundation Engineering”, CRC Press, Taylor & Francis Group, First Indian Reprint, 2010.

R1. Das B M and Sivakugan, N. (2019) Principles of Foundation engineering, Cengage Learning, 9th edition.

R2. Kaniraj, S.R. “Design Aids in Soil Mechanics and Foundation Engineering”, Tata McGraw Hill, 1988.

R3. Gulhati, SK, and Datta, M. “Geotechnical Engineering”, Tata McGraw-Hill Publishing Company Ltd, 2005.

R4. Gopal Ranjan, and A. S. R. Rao, “Basic and Applied Soil Mechanics”, New Age Int. Publishers, 3rd edition, 2015

R5. Relevant BIS, IRC codes and International code of practice

5. Course Plan:

Module	Lecture Session	Lec.	Reference to Ch	Learning Outcome
General requirement for shallow foundations	General principles, concepts, requirement for satisfactory performance of foundations, Types of foundations, selection and their specific applications. Interpretation of geotechnical testing, selecting foundation design parameters from laboratory and field tests	2	NBC (National Building Code)-2026, IS 1904-2021, IS6403-1981 Chapter 5,13 T2, Ch2,3R1, R2,	Design parameters from various tests and general requirements for shallow foundations.



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

Bearing capacity and allowable bearing pressure of shallow foundation	Failure mechanism, generalized bearing capacity equation, local and punching shear failures, corrections for size, shape, depth, water table, compressibility, Effect of load eccentricity and inclination etc., Selection of shear strength parameters, Bearing capacity analysis of footings on layered soils, and slope	4	12T1,6,7, R1,15R4, IS6403, IS8009-part1,	Learning to find Bearing capacity of shallow foundations in different cases
	Different approaches for determination of allowable soil pressure for cohesionless and cohesive soils from lab and field tests (SPT, CPT, PLT etc.), tilt and rotation of footing	4	12T1, IS6403, IS8009-part1,IS1888, R1, R2, notes	Learning procedures to find allowable bearing pressure for shallow foundation
Proportioning of footings	Load combination, Proportioning of isolated and combined footings, subjected to different loads such as vertical, moment, horizontal (single or combined). Optimization, computer applications.	3	14 T1, Class notes	Learning how to proportion shallow foundations
Deep Foundations-bearing capacity and settlement	Types of deep foundations and their applications, selection, general requirements, driven and bored piles, pre-cast and cast in-situ piles, under-reamed piles, Indian case histories.	2	15 T1, IS2911-different parts and sections	Learning Deep Foundations classification, selection, and their applicability and construction
	Load carrying capacity of piles by different methods, pile load tests, settlement for single pile and pile group.	5	IS2911, 15 T1	Learning different methods to find Piles capacity and settlement
Laterally loaded Pile	Various methods for laterally loaded pile analysis	3	16 T1, IS2911, 12R1,	Learning procedures to analyses laterally loaded Pile
Earth pressures theories	Various theories for computation of earth pressures, Earth Pressure theories, Coulomb and Rankine approaches, smooth and rough walls, inclined backfill, depth of tension crack, lateral pressure due to surcharge loads	3	11 T1, IRC:6, 7-R1	Learning different approaches of earth pressures theories
Earth retaining structures	Classification and selection of different type of retaining walls. Analysis of different types of retaining walls, stability condition,	3	17T2, 19 T1, R5, 8-R1	Selection and analysis of retaining walls,
Introduction to geotechnical earthquake engineering	Basics of soil dynamics, Dynamic properties of soil, using laboratory and field tests. seismic design guidelines for foundations and geotechnical structure, liquefaction of soil, screening criterion, evaluation of liquefaction potential.	4	Class notes, IS1893-part1-2016, code of practices, R2, R3, IS 5249	Learning basic concepts of geotechnical earthquake engineering, Evaluation of dynamic Properties of Soil and liquefaction of soils
Machine foundation.	General requirements and design criteria - Stiffness and damping parameters, Analysis and design of machine foundations for reciprocating engines, impact type machines. Limitations of BIS code of practices	2	R2, R3, R4, IS 2974-part1,2	Analysis and design of foundations for different type of machines.



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

Geosynthetic s	Introduction to different type of geosynthetics, and various applications, Advantages and applications of reinforced earth walls, Indian case histories,	2	17 T2, R3, IRC codes, notes	Learning Geosynthetics applications and benefits
Ground improvement techniques	Soil stabilization and ground improvement techniques for difficult or problematic ground conditions, anti-liquefaction measures, Dynamic compaction, Vibro techniques, preloading, vertical drains, PVD, stone columns, grouting etc,	3	Ch 21 T1, 29, 30, 31R3, IRC guidelines, IS 13094, other IS codes and notes	Learning ground improvement techniques principles, advantages, limitations and selection.
Stability of slopes	Stability of slopes, limit equilibrium methods, methods of slices.	3	10 T1,	Methods of Stability of slopes
	Total	43		

6. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Nature of component (Close Book/ Open Book)
Mid	90 Min.	25	AUGSD notices	Closed Book
Comprehensive Examination	180 Min.	45	AUGSD notices	Closed Book
Test/Assignments		30	4 tests (dates to be finalized in the class and will be informed through BITS email)	Out of 4 best 3 scores will be considered. Open Book

7. Consultation Hour: Th6 (or call me/ message @9887692025)

8. Notices: Notices will be sent to your BITS email id only.

9. Make-up Policy: Makeup will be given only for genuine cases (accompanied by proof, with **prior permission**) for Mid-sem. or comprehensive exam only. No makeup for Tests/ assignments (one buffer will be given).

10. Note (if any): Reading assignments will be given whenever necessary.

Instructor-in-charge
Course No.